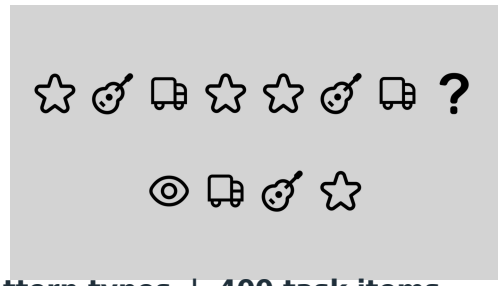
Abstract pattern reasoning in humans and language models

A common task links behavior, model representations and fixation-locked EEG

A One task, two routes to representational geometry



8 pattern types | 400 task items

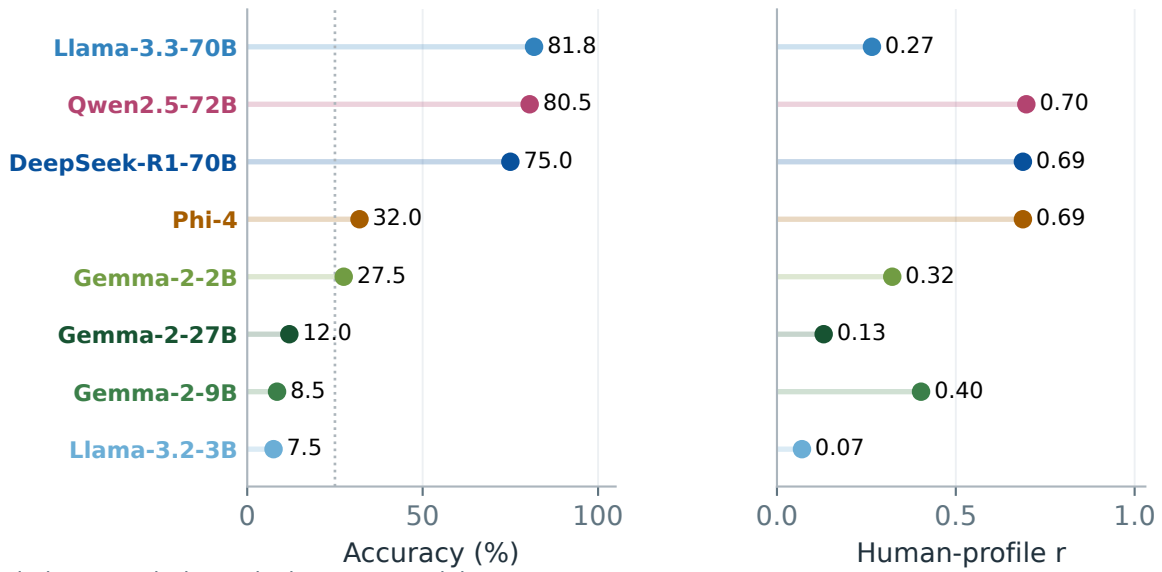
HUMANS
25 participants: behavior
24 matched participants: EEG

LANGUAGE MODELS
8 models; icon-name sequences
Task-selected internal layer

COMPARE PATTERN GEOMETRY
8-pattern RDMs
28 unique pattern pairs
Pearson RSA

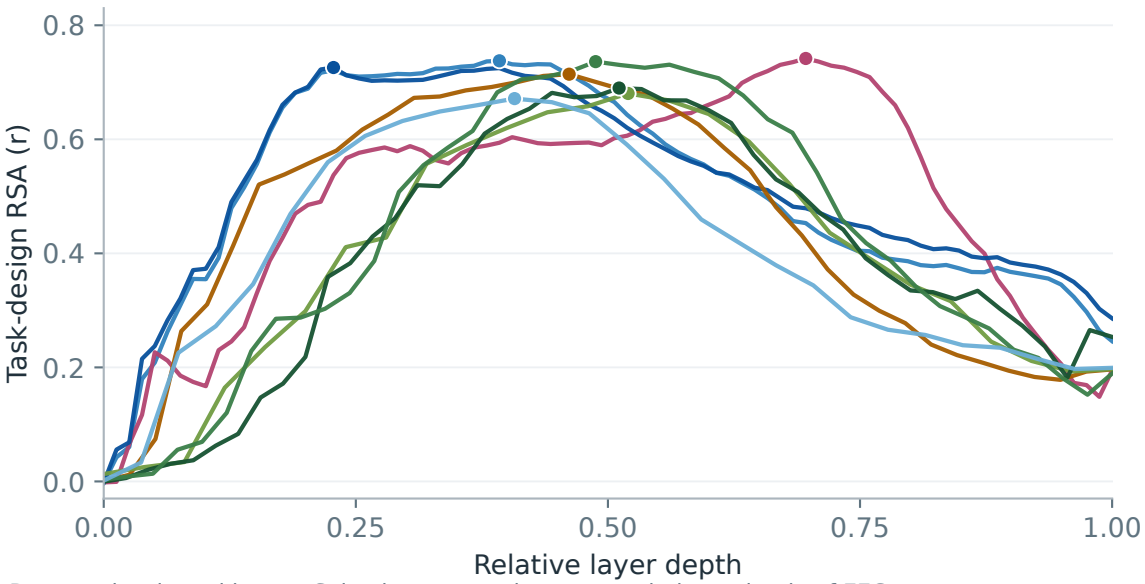
B Behavior: accuracy and human-like difficulty

82.5% human mean **40.6%** model mean



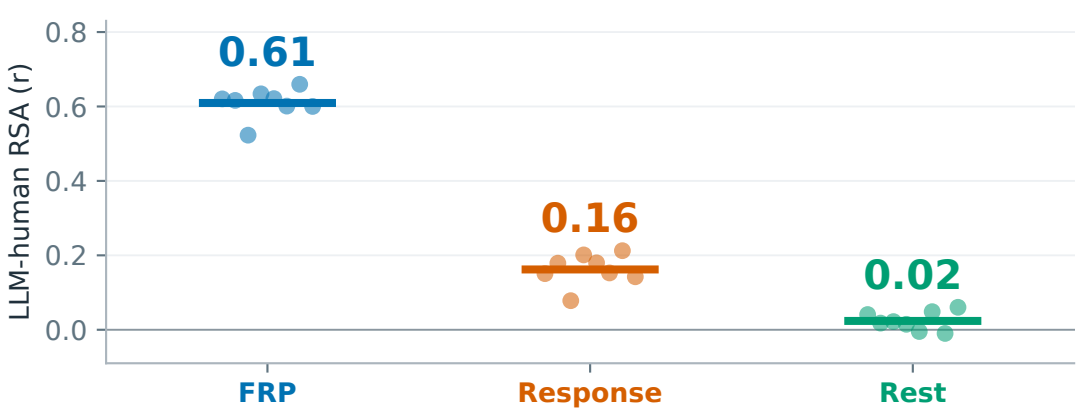
Profile correlations match the rank plot; r across eight patterns.

C Task structure emerges inside the models



Dots mark selected layers. Selection uses task structure, independently of EEG.

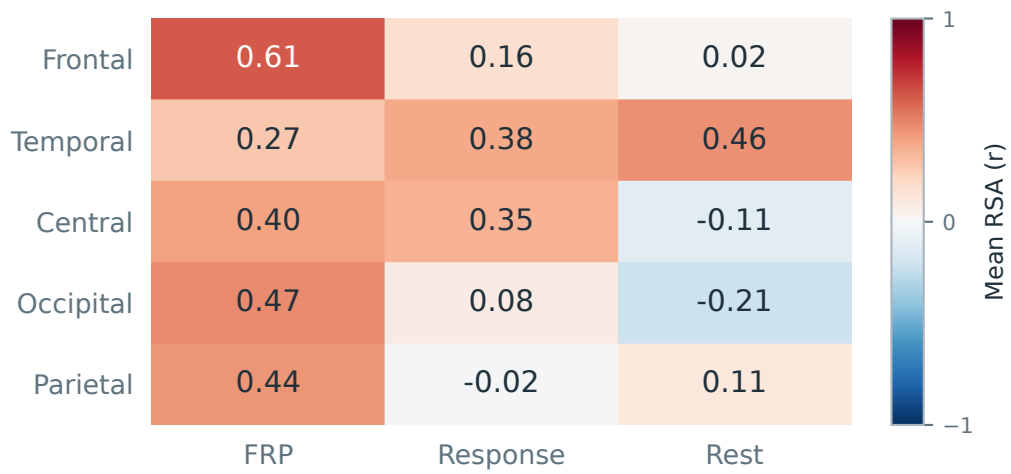
D Frontal correspondence is strongest for FRP



Dots: eight models. Horizontal marks: means. Matched N = 24.

Frontal FRP pattern-label test: p = .010; Holm-adjusted p = .030.

E Regional context matters



Means across eight models; color describes effect size, not significance.